

Jorge Manuel Torre

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Education

Virginia Tech

B.S. Computer Science, Overall GPA: 3.32/4.0, Major GPA: 3.50/4.0

Blacksburg, VA

May 2026

Relevant Coursework: Security of AI, Machine Learning (CS 4824), Linux Kernel Programming, Software Reverse Engineering, Computer Architecture, Computer Systems, Data Structures & Algorithms.

Projects & Experience

CHUD: Catastrophic Harmful Update Detection • GitHub

January 2026 – April 2026

Python, PyTorch, LoRA, Llama-2-7b, Llama-Guard-3-8b, Unsloth

Team of 4 (CS Capstone)

- Investigated weaknesses in the LoX defense (Perin et al., COLM 2025) via an iterative LoRA fine-tuning attack on Llama-2-7b that interleaves benign GSM8k and adversarial BeaverTails samples to drive catastrophic forgetting in safety-critical subspaces.
- Built an Attack Success Rate (ASR) evaluation pipeline using Llama-Guard-3-8b as a judge model, calibrated against AdvBench (99% accuracy, 2% FNR) and BeaverTails; ran 30+ ASR experiments across attack mixes and learning-rate sweeps.
- Identified a robustness gap in LoX: direct harmful fine-tuning achieves 91–96% ASR on Chat-aligned Llama-2, LoX reduces this to 37–63% under direct attack, but iterative mixed-data attack restores 89% ASR — flattening the safety landscape without eliminating exploitability.
- Provisioned dual VT GPU cluster environments (x86-64 and aarch64) for the team's training pipeline; owned QLoRA fine-tuning evaluation and integrated results into the final ASR analysis.

Celeste-RL: Deep Reinforcement Learning for a Precision Platformer • GitHub • Website

January 2026 – April 2026

Python, PyTorch, NumPy, matplotlib, CUDA

- Conducted a controlled comparison of five RL methods (DQN, Behavioral Cloning, Hybrid, Curriculum, Dueling DQN with curiosity) trained to play Celeste Classic room 0 on the Pyleste emulator.
- Identified state representation as the dominant factor over algorithm choice: replacing raw PICO-8 tile IDs with a 5-class semantic encoding (air / solid / spike / out-of-bounds / other) raised plain DQN completion from 8% to 57–68% with no other change.
- Shipped a fully reproducible pipeline with batched evaluation, automated comparison plots, side-by-side gameplay GIF generator, GitHub-Pages-deployed results writeup, and a published debugging devlog covering 9 iteration phases.

Garage64: Custom Entity Model Studio • GitHub

August 2025 – Present

React, Three.js, Django, SQLite, Docker

- Engineered a full-stack web tool with a real-time Three.js 3D workspace, Django REST API, and a one-click `.jem/.jpm` export pipeline — demonstrates production engineering breadth alongside the ML work.

Experimental Home Lab

January 2021 – Present

Proxmox, OPNsense, NixOS, Docker, CUDA

- Architected a multi-VM Proxmox lab with reproducible NixOS / Docker workspaces and a CUDA-enabled training environment used for the ML projects above.

Leadership & Community

Team Lead, Database Management Systems (CS 4604)

May 2025 – August 2025

- Directed a 3-person team in collaborative schema design and SQL development, facilitating design discussions and integrating individual contributions into a unified submission.

Team Lead, Intermediate Software Design (CS 3704)

January 2025 – April 2025

- Led a 5-person team through the full software lifecycle including requirements gathering, sprint planning, and code reviews; coordinated deliverables across team members with differing schedules and skill levels.

Garage32 (Community Project)

March 2020 – December 2021

- Led development and distribution of a graphical asset pack for the Minecraft community, reaching **600+ downloads**; promotional content earned **30K+ likes** on a single TikTok post.

Skills

ML / AI: PyTorch, LoRA, Fine-Tuning, LLM Safety & Alignment, Reinforcement Learning (DQN, BC, Curriculum), Llama / Llama-Guard, Adversarial ML, Research Methodology

Languages: Python, C, C++, Java, SQL, JavaScript, Assembly (ARM, x86-64, RISC-V), Bash

Infrastructure & Tooling: CUDA, Docker, NixOS, Proxmox, Linux, Git, CI/CD

Also familiar with: React, Three.js, Django, PostgreSQL